Menu

NORMALIZATION: per_kernel

GCD: ALL

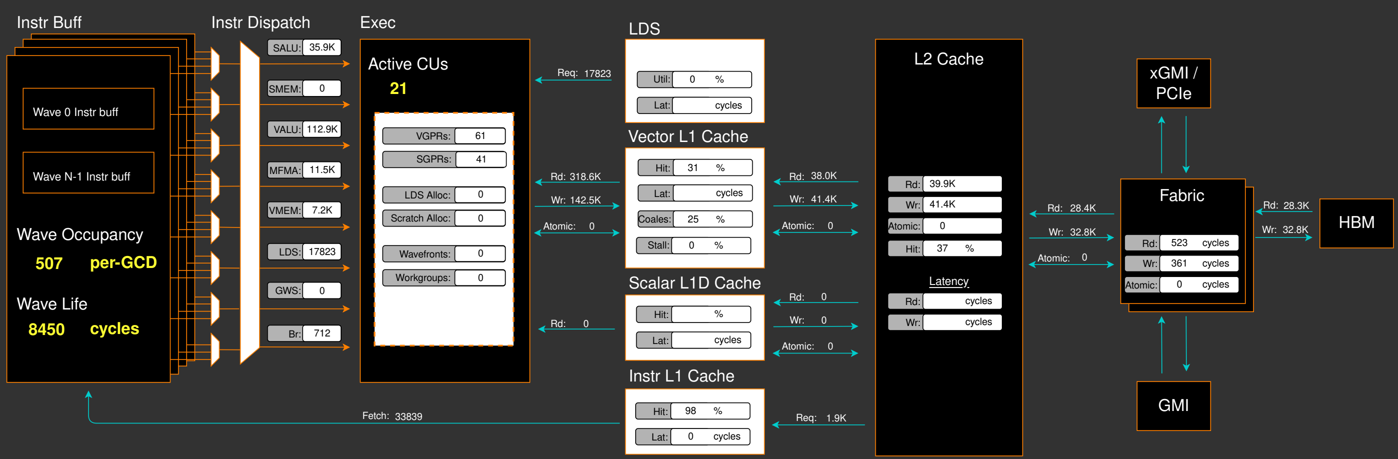
DISPATCH FILTER: ALL

TOP N: 10

KERNELS: × bias_kernel × matmul_kernel

Report Bug

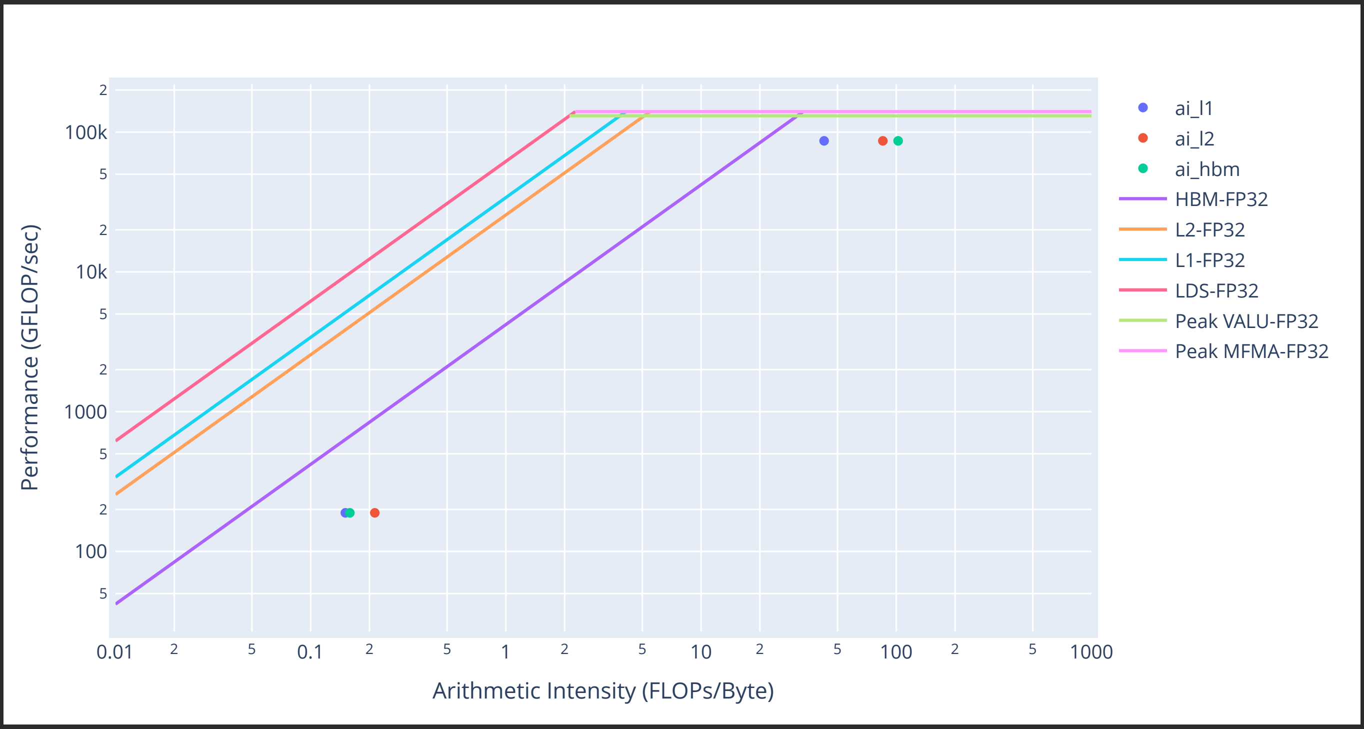
Placeholder. Guided Analysis coming soon...



Menu NORMALIZATION: per_kernel GCD: ALL DISPATCH FILTER: ALL

TOP N: 10 KERNELS: × bias_kernel × matmul_kernel Report Bug

Empirical Roofline Analysis (Flops)



Menu

NORMALIZATION: per_kernel GCD: ALL DISPATCH FILTER: ALL TOP N: 10

KERNELS: × bias_kernel × matmul_kernel

Report Bug

2. System Speed-of-Light

2.1 Speed-of-Light

Metric	Avg	Unit	Peak	Pct of Peak
VALU FLOPs	177.30	Gflop/s	81715.20	0.22
VALU IOPs	625.76	Giop/s	81715.20	0.77
MFMA FLOPs (F8)	0.00	Gflop/s	2614886.40	0.00
MFMA FLOPs (BF16)	0.00	Gflop/s	1307443.20	0.00
MFMA FLOPs (F16)	7701.98	Gflop/s	1307443.20	0.59
MFMA FLOPs (F32)	0.00	Gflop/s	163430.40	0.00
MFMA FLOPs (F64)	0.00	Gflop/s	163430.40	0.00
MFMA FLOPs (F6F4)		Gflop/s		
MFMA IOPs (Int8)	0.00	Giop/s	2614886.40	0.00
Active CUs	21.00	Cus	304.00	6.91
SALU Utilization	0.40	Pct	100.00	0.40
VALU Utilization	1.14	Pct	100.00	1.14
MFMA Utilization	0.45	Pct	100.00	0.45
VMEM Utilization	0.07	Pct	100.00	0.07
Branch Utilization	0.01	Pct	100.00	0.01
VALU Active Threads	63.44	Threads	64.00	99.12
IPC	0.43	Instr/cycle	5.00	8.52
Wavefront Occupancy	10656.55	Wavefronts	9728.00	109.55
Theoretical LDS Bandwidth	485.66	Gb/s	81715.20	0.59
LDS Bank Conflicts/Access	0.72	Conflicts/access	32.00	2.26
vL1D Cache Hit Rate	31.06	Pct	100.00	31.06
vL1D Cache BW	2692.31	Gb/s	81715.20	3.29
L2 Cache Hit Rate	37.15	Pct	100.00	37.15
L2 Cache BW	1877.30	Gb/s	34406.40	5.46
L2-Fabric Read BW	622.97	Gb/s	5324.80	11.70
L2-Fabric Write BW	552.93	Gb/s	5324.80	10.38
L2-Fabric Read Latency	523.35	Cycles		
L2-Fabric Write Latency	360.73	Cycles		
sL1D Cache Hit Rate		Pct	100.00	
sL1D Cache BW	0.00	Gb/s	21504.00	0.00
L1I Hit Rate	99.72	Pct	100.00	99.72
L1I BW	330.77	Gb/s	21504.00	1.54
L1I Fetch Latency		Cycles		

Menu NORMALIZATION: per_kernel GCD: ALL DISPATCH FILTER: ALL TOP N: 10

KERNELS: × bias_kernel × matmul_kernel

Report Bug

5. Command Processor (CPC/CPF)

5.1 Command Processor Fetcher

Metric	Avg	Min	Max	Unit
CPF Utilization	100.00	100.00	100.00	Pct
CPF Stall	3.49	0.09	7.25	Pct
CPF-L2 Utilization	35.61	4.14	48.39	Pct
CPF-L2 Stall	0.00	0.00	0.00	Pct
CPF-UTCL1 Stall	3.49	0.09	7.25	Pct

5.2 Packet Processor

Metric	Avg	Min	Max	Unit
CPC SYNC FIFO Full Rate				Pct
CPC CANE Stall Rate				Pct
CPC ADC Utilization				Pct
CPC Utilization	100.00	100.00	100.00	Pct
CPC Stall Rate	34.63	24.56	43.79	Pct
CPC Packet Decoding Utilization	77.49	49.90	105.19	Pct
CPC-Workgroup Manager Utilization	1.98	0.56	2.61	Pct
CPC-L2 Utilization	1.04	0.28	1.31	Pct
CPC-UTCL1 Stall	0.63	0.17	6.96	Pct
CPC-UTCL2 Utilization	0.60	0.16	4.86	Pct

6. Workgroup Manager (SPI)

6.1 Workgroup Manager Utilizations

Metric	Avg	Min	Max	Unit
Schedule-Pipe Wave Occupancy				Wave
Accelerator Utilization	100.00	100.00	100.00	Pct
Scheduler-Pipe Utilization	0.00	0.00	0.00	Pct
Scheduler-Pipe Wave Utilization				Pct
Workgroup Manager Utilization	33.38	18.48	80.92	Pct
Shader Engine Utilization	25.69	12.69	78.35	Pct
SIMD Utilization	4.28	1.58	16.86	Pct
Dispatched Workgroups	0.00	0.00	0.00	Workgroups
Dispatched Wavefronts	0.00	0.00	0.00	Wavefronts
VGPR Writes				Cycles/wave
SGPR Writes				Cycles/wave

6.2 Workgroup Manager - Resource Allocation

Metric	Avg	Min	Max	Unit
Not-scheduled Rate (Workgroup Manager)	0.00	0.00	0.00	Pct
Not-scheduled Rate (Scheduler-Pipe)	0.00	0.00	0.00	Pct
Scheduler-Pipe FIFO Full Rate				Pct
Scheduler-Pipe Stall Rate	0.00	0.00	0.00	Pct
Scratch Stall Rate	0.00	0.00	0.00	Pct
Insufficient SIMD Waveslots	0.00	0.00	0.00	Pct
Insufficient SIMD VGPRs	0.00	0.00	0.00	Pct
Insufficient SIMD SGPRs	0.00	0.00	0.00	Pct
Insufficient CU LDS	0.00	0.00	0.00	Pct
Insufficient CU Barriers	0.00	0.00	0.00	Pct
Reached CU Workgroup Limit	0.00	0.00	0.00	Pct
Reached CU Wavefront Limit	0.00	0.00	0.00	Pct

Menu NORMALIZATION: per_kernel GCD: ALL DISPATCH FILTER: ALL TOP N: 10

KERNELS: × bias_kernel × matmul_kernel Report Bug

7. Wavefront

7.1 Wavefront Launch Stats

Metric	Avg	Min	Max	Unit
Grid Size	19512.63	8192.00	32768.00	Work items
Workgroup Size	345.07	256.00	512.00	Work items
Total Wavefronts	0.00	0.00	0.00	Wavefronts
Saved Wavefronts	0.00	0.00	0.00	Wavefronts
Restored Wavefronts	0.00	0.00	0.00	Wavefronts
VGPRs	61.03	16.00	104.00	Registers
AGPRs	2.21	0.00	192.00	Registers
SGPRs	40.65	32.00	48.00	Registers
LDS Allocation	0.00	0.00	0.00	Bytes
Scratch Allocation	0.00	0.00	0.00	Bytes/workitem

7.2 Wavefront Runtime Stats

Metric	Avg	Min	Max	Unit
Kernel Time (Nanosec)	5418.03	2986.75	43268.25	Ns
Kernel Time (Cycles)	27423.58	19622.38	351674.38	Cycle
Instructions per wavefront	572.94	247.00	3555.00	Instr/ wavefront
Wave Cycles	3078498.74	709024.00	24266004.00	Cycles per kernel
Dependency Wait Cycles	1543702.59	291304.00	68019812.00	Cycles per kernel
Issue Wait Cycles	751289.76	80192.00	8404596.00	Cycles per kernel
Active Cycles	766412.04	310784.00	4766232.00	Cycles per kernel
Wavefront Occupancy	10656.55	29.14	33580.72	Wavefronts

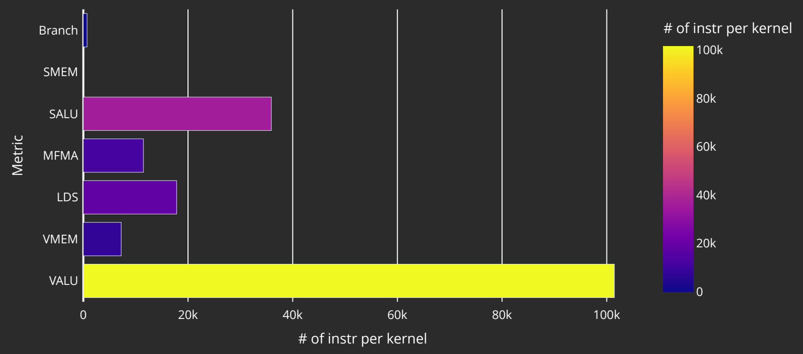
Menu NORMALIZATION: per_kernel GCD: ALL DISPATCH FILTER: ALL TOP N: 10

KERNELS: × bias_kernel × matmul_kernel

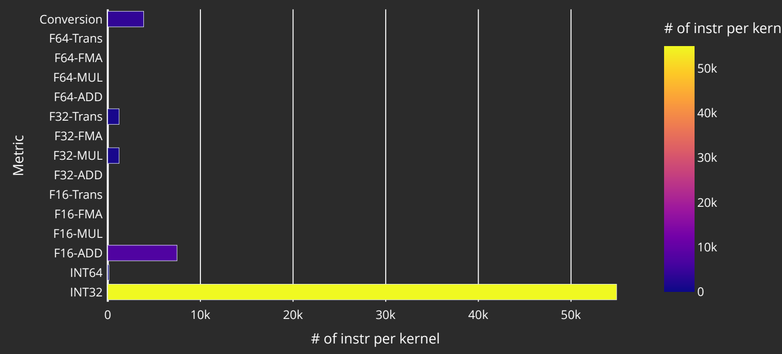
Report Bug

10. Compute Units - Instruction Mix

10.1 Overall Instruction Mix



10.2 VALU Arithmetic Instr Mix



10.3 VMEM Instr Mix

Metric	Avg	Min	Max	Unit
Global/Generic Instr	0.00	0.00	0.00	Instr per kernel
Global/Generic Read	0.00	0.00	0.00	Instr per kernel
Global/Generic Write	0.00	0.00	0.00	Instr per kernel
Global/Generic Atomic	0.00	0.00	0.00	Instr per kernel
Spill/Stack Instr	7228.09	4224.00	36864.00	Instr per kernel
Spill/Stack Coalesceable Instr				Instr per kernel
Spill/Stack Read	5000.77	2176.00	32768.00	Instr per kernel
Spill/Stack Write	2227.32	2048.00	4096.00	Instr per kernel
Spill/Stack Atomic	0.00	0.00	0.00	Instr per kernel

10.4 MFMA Arithmetic Instr Mix

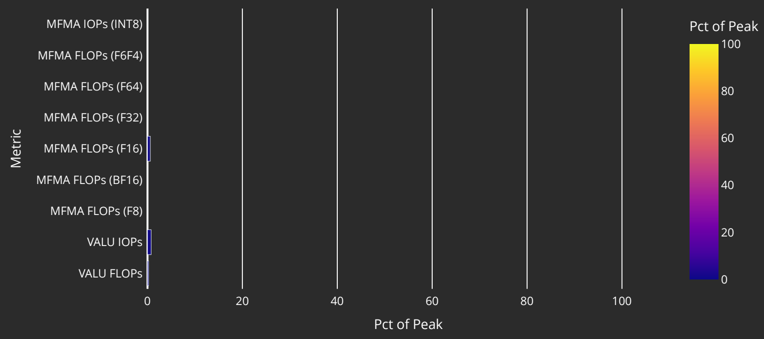
Metric	Avg	Min	Max	Unit
MFMA-I8	0.00	0.00	0.00	Instr per kernel
MFMA-F8	0.00	0.00	0.00	Instr per kernel
MFMA-F16	11476.35	0.00	131072.00	Instr per kernel
MFMA-BF16	0.00	0.00	0.00	Instr per kernel
MFMA-F32	0.00	0.00	0.00	Instr per kernel
MFMA-F64	0.00	0.00	0.00	Instr per kernel
MFMA-F6F4				Instr per kernel

Menu NORMALIZATION: per_kernel GCD: ALL DISPATCH FILTER: ALL TOP N: 10

KERNELS: × bias_kernel × matmul_kernel Report Bug

11. Compute Units - Compute Pipeline

11.1 Speed-of-Light



11.2 Pipeline Stats

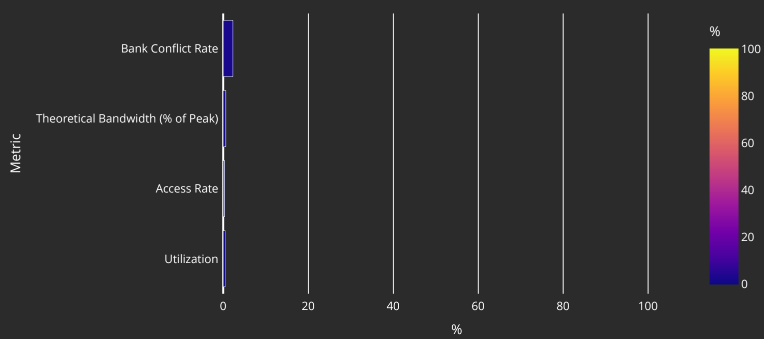
Metric	Avg	Min	Max	Unit
IPC	0.43	0.21	0.61	Instr/cycle
IPC (Issued)	0.96	0.85	0.98	Instr/cycle
SALU Utilization	0.40	0.07	0.79	Pct
VALU Utilization	1.14	0.55	3.46	Pct
VALU Co-Issue Efficiency				Pct
VMEM Utilization	0.07	0.03	0.20	Pct
Branch Utilization	0.01	0.00	0.13	Pct
VALU Active Threads	63.44	63.00	63.94	Threads
MFMA Utilization	0.45	0.00	5.68	Pct
MFMA Instr Cycles	32.00	32.00	32.00	Cycles/instr
VMEM Latency				Cycles
SMEM Latency				Cycles

11.3 Arithmetic Operations

Metric	Avg	Min	Max	Unit
FLOPs (Total)	188663005.79	589824.00	2147745792.00	Ops per kernel
IOPs (Total)	3523673.66	2015232.00	18776064.00	Ops per kernel
F8 OPs	0.00	0.00	0.00	Ops per kernel
F16 OPs	188506904.77	524288.00	2147483648.00	Ops per kernel
BF16 OPs	0.00	0.00	0.00	Ops per kernel
F32 OPs	156101.01	65536.00	262144.00	Ops per kernel
F64 OPs	0.00	0.00	0.00	Ops per kernel
F6F4 OPs				Ops per kernel
INT8 OPs	0.00	0.00	0.00	Ops per kernel

12. Local Data Share (LDS)

12.1 Speed-of-Light



12.2 LDS Stats

Metric	Avg	Min	Max	Unit
LDS Instrs	17822.60	0.00	212992.00	Instr per kernel
LDS LOAD				Instr per kernel
LDS STORE				Instr per kernel
LDS ATOMIC				Instr per kernel
LDS LOAD Bandwidth				Gbps
LDS STORE Bandwidth				Gbps
LDS ATOMIC Bandwidth				Gbps
Theoretical Bandwidth	11432993.03	0.00	134217728.00	Bytes per kernel
LDS Latency				Cycles
Bank Conflicts/Access	0.72	0.00	1.00	Conflicts/access
Index Accesses	89849.81	0.00	1048576.00	Cycles per kernel
Atomic Return Cycles	0.00	0.00	0.00	Cycles per kernel
Bank Conflict	529.55	0.00	2304.00	Cycles per kernel
Addr Conflict	0.00	0.00	0.00	Cycles per kernel
Unaligned Stall	0.00	0.00	0.00	Cycles per kernel
Mem Violations	0.00	0.00	0.00	Accesses per kernel
LDS Command FIFO Full Rate				Cycles per kernel
LDS Data FIFO Full Rate				Cycles per kernel

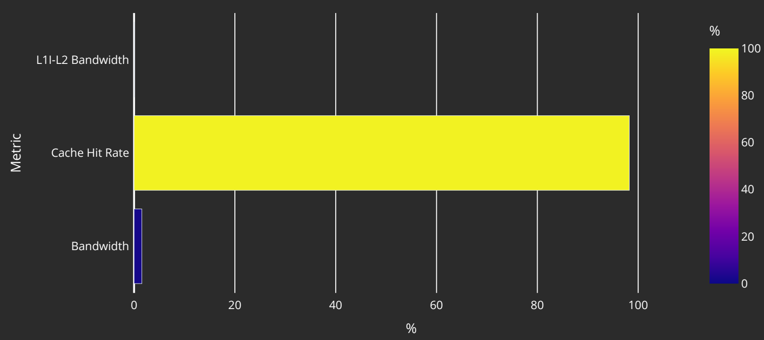
Menu NORMALIZATION: per_kernel GCD: ALL DISPATCH FILTER: ALL TOP N: 10

KERNELS: bias_kernel matmul_kernel

Report Bug

13. Instruction Cache

13.1 Speed-of-Light



13.2 Instruction Cache Accesses

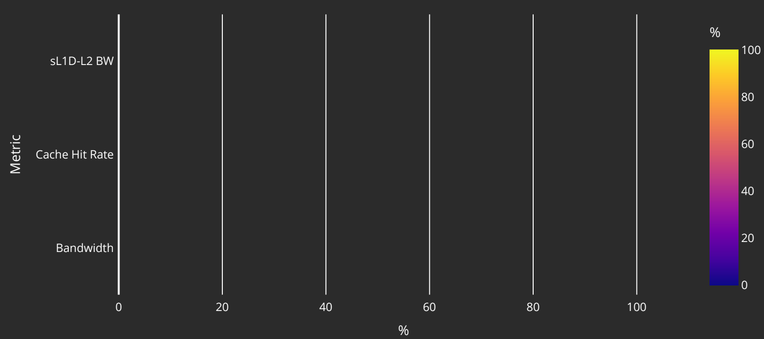
Metric	Avg	Min	Max	Unit
Req	33839.04	14080.00	217088.00	Req per kernel
Hits	33472.77	13766.00	216303.00	Hits per kernel
Misses - Non Duplicated	53.67	32.00	162.00	Misses per kernel
Misses - Duplicated	312.31	192.00	804.00	Misses per kernel
Cache Hit Rate	98.25	96.98	99.65	Pct
Instruction Fetch Latency				Cycles

13.3 Instruction Cache - L2 Interface

Metric	Avg	Min	Max	Unit
L1I-L2 Bandwidth	120947.02	65536.00	535040.00	Bytes per kernel

14. Scalar L1 Data Cache

14.1 Speed-of-Light



14.2 Scalar L1D Cache Accesses

Metric	Avg	Min	Max	Unit
Req	0.00	0.00	0.00	Req per kernel
Hits	0.00	0.00	0.00	Req per kernel
Misses - Non Duplicated	0.00	0.00	0.00	Req per kernel
Misses- Duplicated	0.00	0.00	0.00	Req per kernel
Cache Hit Rate				Pct
Read Req (Total)	0.00	0.00	0.00	Req per kernel
Atomic Req	0.00	0.00	0.00	Req per kernel
Read Req (1 DWord)	0.00	0.00	0.00	Req per kernel
Read Req (2 DWord)	0.00	0.00	0.00	Req per kernel
Read Req (4 DWord)	0.00	0.00	0.00	Req per kernel
Read Req (8 DWord)	0.00	0.00	0.00	Req per kernel
Read Req (16 DWord)	0.00	0.00	0.00	Req per kernel

14.3 Scalar L1D Cache - L2 Interface

Metric	Avg	Min	Max	Unit
sL1D-L2 BW	0.00	0.00	0.00	Bytes per kernel
Read Req	0.00	0.00	0.00	Req per kernel
Write Req	0.00	0.00	0.00	Req per kernel
Atomic Req	0.00	0.00	0.00	Req per kernel
Stall Cycles	0.00	0.00	0.00	Cycles per kernel

Menu NORMALIZATION: per_kernel GCD: ALL DISPATCH FILTER: ALL TOP N: 10

KERNELS: × bias_kernel × matmul_kernel

Report Bug

15. Address Processing Unit and Data Return Path (TA/TD)

15.1 Address Processing Unit

Metric	Avg	Min	Max	Unit
Address Processing Unit Busy	2.22	0.86	8.80	Pct
Address Stall	0.58	0.00	5.81	Pct
Data Stall	0.56	0.01	2.91	Pct
Data-Processor → Address Stall	0.00	0.00	0.00	Pct
Sequencer → TA Address Stall				Cycles per kernel
Sequencer → TA Command Stall				Cycles per kernel
Sequencer → TA Data Stall				Cycles per kernel
Total Instructions	7228.09	4224.00	36864.00	Instructions per kernel
Global/Generic Instructions	0.00	0.00	0.00	Instructions per kernel
Global/Generic Read Instructions	0.00	0.00	0.00	Instructions per kernel
Global/Generic Read Instructions for LDS				Instructions per kernel
Global/Generic Write Instructions	0.00	0.00	0.00	Instructions per kernel
Global/Generic Atomic Instructions	0.00	0.00	0.00	Instructions per kernel
Spill/Stack Instructions	7228.09	4224.00	36864.00	Instructions per kernel
Spill/Stack Read Instructions	5000.77	2176.00	32768.00	Instructions per kernel
Spill/Stack Read Instructions for LDS				Instructions per kernel
Spill/Stack Write Instructions	2227.32	2048.00	4096.00	Instructions per kernel
Spill/Stack Atomic Instructions	0.00	0.00	0.00	Instructions per kernel
Spill/Stack Total Cycles	115296.44	66048.00	589824.00	Cycles per kernel
Spill/Stack Coalesced Read	117.68	0.00	512.00	Cycles per kernel
Spill/Stack Coalesced Write	0.00	0.00	0.00	Cycles per kernel

15.2 Data-Return Path

Metric	Avg	Min	Max	Unit
Data-Return Busy	2.96	1.06	18.52	Pct
Cache RAM → Data-Return Stall	2.18	0.64	17.99	Pct
Workgroup manager → Data-Return Stall	0.00	0.00	0.00	Pct
Coalescable Instructions	29.42	0.00	128.00	Instructions per kernel
Read Instructions	5000.77	2176.00	32768.00	Instructions per kernel
Write Instructions	2227.32	2048.00	4096.00	Instructions per kernel
Atomic Instructions	0.00	0.00	0.00	Instructions per kernel
Write Ack Instructions				Instructions per kernel

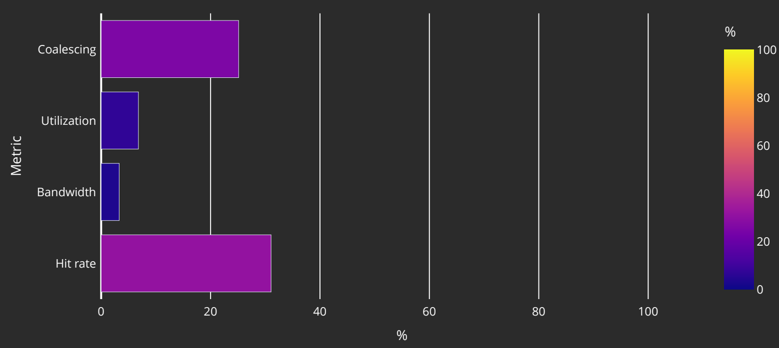
Menu NORMALIZATION: per_kernel GCD: ALL DISPATCH FILTER: ALL TOP N: 10

KERNELS: bias_kernel matmul_kernel

Report Bug

16. Vector L1 Data Cache

16.1 Speed-of-Light



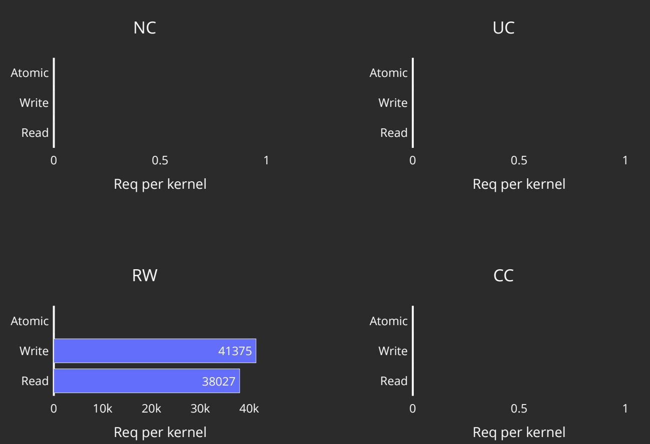
16.2 L1D Cache Stalls (%)

Metric	Min	Q1	Median	Q3	Max
Stalled on L2 Data	1.49	2.71	4.14	4.69	14.18
Stalled on L2 Req	0.00	0.00	0.00	0.00	0.01
Stalled on Address					
Stalled on Data					
Stalled on Latency FIFO					
Stalled on Request FIFO					
Stalled on Read Return					
Tag RAM Stall (Read)	0.00	0.00	0.00	0.00	0.00
Tag RAM Stall (Write)	0.00	0.00	0.00	0.00	1.40
Tag RAM Stall (Atomic)	0.00	0.00	0.00	0.00	0.00

16.3 L1D Cache Accesses

Metric	Avg	Min	Max	Unit
Total Req	461185.77	264192.00	2359296.00	Req per kernel
Read Req	318637.42	133120.00	2097152.00	Req per kernel
Write Req	142548.35	131072.00	262144.00	Req per kernel
Atomic Req	0.00	0.00	0.00	Req per kernel
Cache BW	16961403.87	8454144.00	100663296.00	Bytes per kernel
Cache Hit Rate	31.06	25.39	50.00	Pct
Cache Accesses	132510.97	66048.00	786432.00	Req per kernel
Cache Hits	53109.74	16768.00	393216.00	Req per kernel
Invalidations	51.43	32.00	64.00	Req per kernel
L1-L2 BW	7515340.39	4210688.00	41943040.00	Bytes per kernel
Tag RAM 0 Req				Req per kernel
Tag RAM 1 Req				Req per kernel
Tag RAM 2 Req				Req per kernel
Tag RAM 3 Req				Req per kernel
L1-L2 Read	38025.97	16512.00	262144.00	Req per kernel
L1-L2 Write	41375.26	32768.00	131072.00	Req per kernel
L1-L2 Atomic	0.00	0.00	0.00	Req per kernel
L1 Access Latency				Cycles
L1-L2 Read Latency				Cycles
L1-L2 Write Latency				Cycles

16.4 L1D - L2 Transactions



16.5 L1D Addr Translation

Metric	Avg	Min	Max	Units
Req	132510.97	66048.00	786432.00	Req per kernel
Inflight Req				Req per kernel
Hit Ratio	96.80	93.93	100.00	Pct
Hits	130243.79	64512.00	786432.00	Req per kernel
Translation Misses	125.59	0.00	192.00	Req per kernel
Misses under Translation Miss				Req per kernel
Permission Misses	0.07	0.00	128.00	Req per kernel

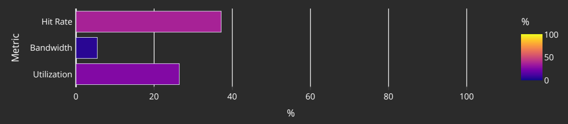
16.6 L1D Addr Translation Stalls

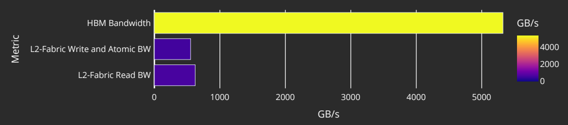
Metric	Avg	Min	Max	Units
Cache Full Stall				Cycles per kernel
Cache Miss Stall				Cycles per kernel
Serialization Stall				Cycles per kernel
Thrashing Stall				Cycles per kernel
Latency FIFO Stall				Cycles per kernel
Resident Page Full Stall				Cycles per kernel
UTCL2 Stall				Cycles per kernel

Menu NORMALIZATION: per_kernel GCD: ALL DISPATCH FILTER: ALL TOP N: 10 KERNELS: × bias_kernel × matmul_kernel Report Bug

17. L2 Cache

17.1 Speed-of-Light





17.2 L2 - Fabric Transactions

Metric	Avg	Min	Max	Unit
Read BW	3605101.20	2129920.00	18953216.00	Bytes per kernel
HBM Read Traffic	99.98	99.90	100.10	Pct
Remote Read Traffic	0.03	0.00	0.10	Pct
Uncached Read Traffic	0.08	0.00	0.48	Pct
Write and Atomic BW	2097152.00	2097152.00	2097152.00	Bytes per kernel
HBM Write and Atomic Traffic	100.00	100.00	100.00	Pct
Remote Write and Atomic Traffic	0.00	0.00	0.00	Pct
Atomic Traffic	0.00	0.00	0.00	Pct
Uncached Write and Atomic Traffic	0.00	0.00	0.00	Pct
Read Latency	523.35	492.84	701.25	Cycles
Write and Atomic Latency	360.73	316.07	517.78	Cycles
Atomic Latency				Cycles
Read Stall				Pct
Write Stall				Pct

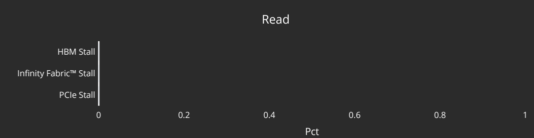
17.3 L2 Cache Accesses

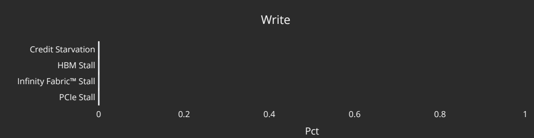
Metric	Avg	Min	Max	Unit
Bandwidth	10407064.48	6438912.00	51423232.00	Bytes per kernel
Read Bandwidth				Bytes per kernel
Write Bandwidth				Bytes per kernel
Atomic Bandwidth				Bytes per kernel
Req	81305.19	50304.00	401744.00	Req per kernel
Read Req	39921.67	17536.00	270504.00	Req per kernel
Write Req	41375.26	32768.00	131072.00	Req per kernel
Atomic Req	0.00	0.00	0.00	Req per kernel
Streaming Req	0.00	0.00	0.00	Req per kernel
Bypass Req				Req per kernel
Probe Req	0.00	0.00	0.00	Req per kernel
Input Buffer Req				Req per kernel
Cache Hit	37.15	34.35	59.06	Pct
Hits	36752.67	17280.00	237280.00	Hits per kernel
Misses	44546.75	33024.00	164464.00	Misses per kernel
Writeback	16496.48	16496.00	16504.00	Cachelines per kernel
Writeback (Internal)	0.00	0.00	0.00	Cachelines per kernel
Writeback (vL1D Req)	16384.00	16384.00	16384.00	Cachelines per kernel
Evict (Internal)	0.00	0.00	0.00	Cachelines per kernel
Evict (vL1D Req)	0.00	0.00	0.00	Cachelines per kernel
NC Req	0.00	0.00	0.00	Req per kernel
UC Req	2.29	0.00	17.00	Req per kernel
CC Req	0.00	0.00	0.00	Req per kernel
RW Req	81291.03	50304.00	401576.00	Req per kernel

17.4 L2 Cache Stalls

Metric	Avg	Min	Max	Unit
Stalled on Latency FIFO				Cycles per kernel
Stalled on Write Data FIFO				Cycles per kernel
Input Buffer Stalled on L2				Cycles per kernel

17.5 L2 - Fabric Interface Stalls





17.6 L2 - Fabric Detailed Transaction Breakdown

Metric	Avg	Min	Max	Unit
Read (32B)	0.00	0.00	0.00	Req per kernel
Read (64B)	375.42	256.00	1232.00	Req per kernel
Read (128B)	27977.14	16512.00	147456.00	Req per kernel
Read (Uncached)	14.99	0.00	80.00	Req per kernel
HBM Read	28348.62	16768.00	148689.00	Req per kernel
Remote Read	4.71	0.00	16.00	Req per kernel
Read Bandwidth - PCIe				Bytes per kernel
Read Bandwidth - Infinity Fabric™				Bytes per kernel
Read Bandwidth - HBM				Bytes per kernel
Write and Atomic (32B)	0.00	0.00	0.00	Req per kernel
Write and Atomic (Uncached)	0.00	0.00	0.00	Req per kernel
Write and Atomic (64B)	32768.00	32768.00	32768.00	Req per kernel
HBM Write and Atomic	32768.00	32768.00	32768.00	Req per kernel
Remote Write and Atomic	0.00	0.00	0.00	Req per kernel
Write Bandwidth - PCIe				Bytes per kernel
Write Bandwidth - Infinity Fabric™				Bytes per kernel
Write Bandwidth - HBM				Bytes per kernel
Atomic	0.00	0.00	0.00	Req per kernel
Atomic - HBM				Req per kernel
Atomic Bandwidth - PCIe				Bytes per kernel
Atomic Bandwidth - Infinity Fabric™				Bytes per kernel
Atomic Bandwidth - HBM				Bytes per kernel